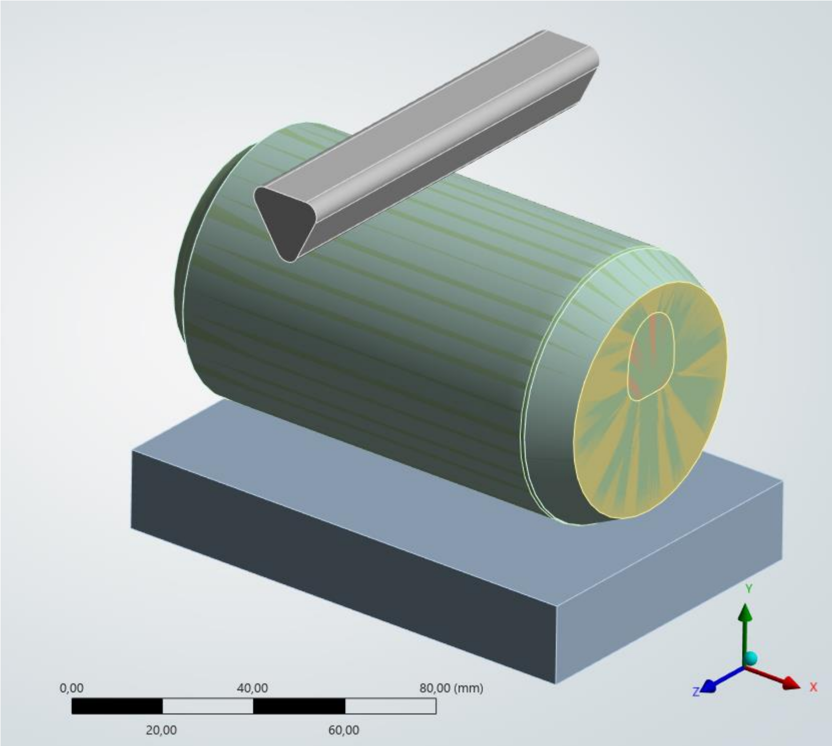
High-Speed Impact & FSI Simulation of a Fluid-Filled Vessel

Project Objective

To simulate a high-velocity impact and catastrophic rupture of a fluid-filled thin-walled aluminum vessel. The primary goal was to analyze non-linear material failure, severe plastic deformation crumpling, and Fluid-Structure Interaction FSI using ANSYS Explicit Dynamics.



Setup & Boundary Conditions

The simulation was configured to replicate a dynamic drop/crush test within a 1 millisecond time frame.

Material Failure Model

The aluminum shell was assigned Bilinear Isotropic Hardening. To simulate realistic tearing, a Plastic Strain Failure criterion was implemented, automatically deleting elements upon reaching a 10% 0.1 Maximum Equivalent Plastic Strain.

Fluid-Structure Interaction FSI

The internal fluid Water was modeled using an Eulerian Virtual reference frame, while the solid structure utilized a standard Lagrangian approach.

Tooling

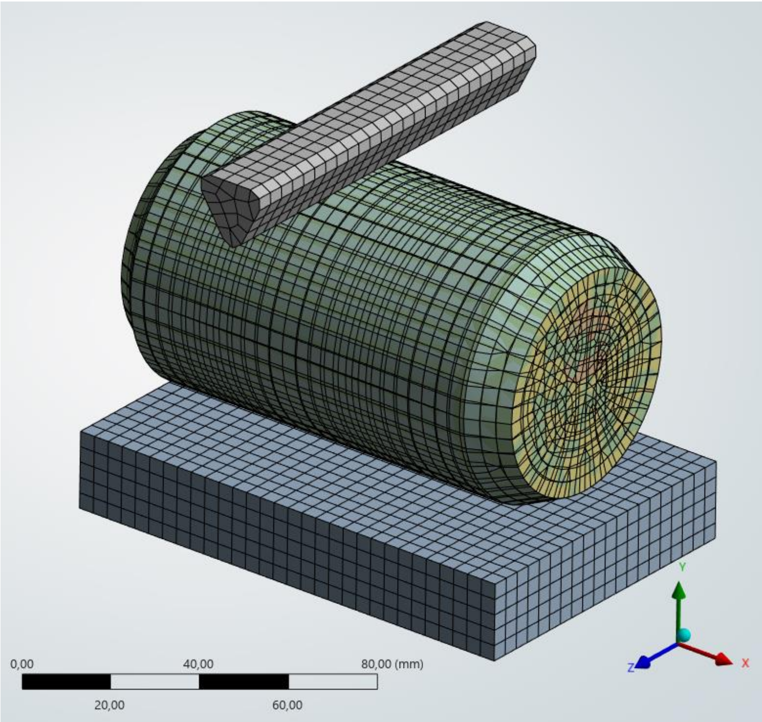
The impact punch and bottom die were modeled as Rigid bodies Structural Steel to optimize solver time.

Contact Formulation

Applied Penalty formulation for body interactions with a Shell Thickness Factor of 1.0 to accurately capture complex self-contact and folding of the thin walls during the crush.

Dynamic Loading & Constraints

The bottom die was rigidly secured Fixed Support, while the impact punch was driven by a non-linear, multi-step displacement profile Tabular Data. This custom time-history loading accurately replicated the high-speed crush and subsequent rebound phase within a strict 1 ms timeframe.



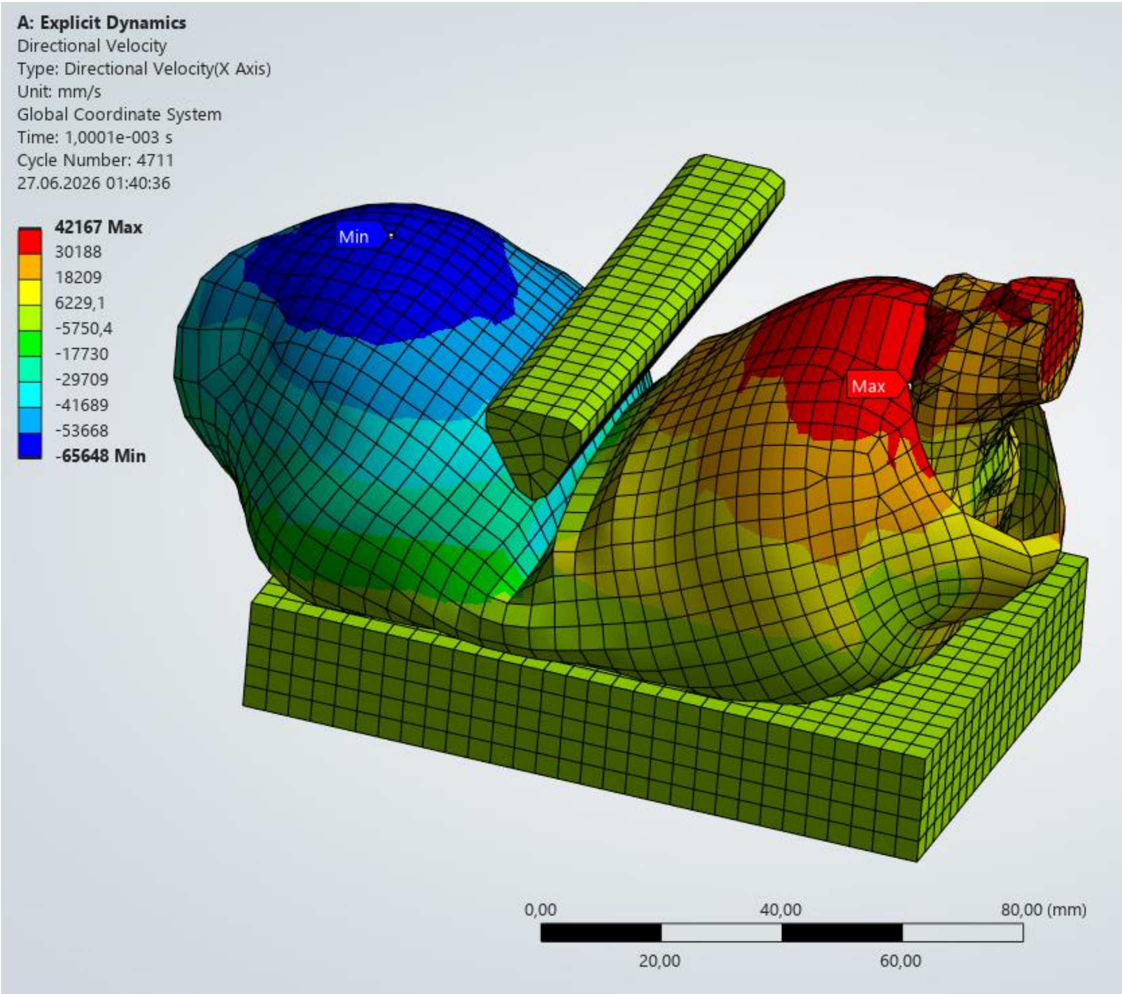
Results & Engineering Conclusion

Rupture Dynamics

The solver successfully captured the exact moment of structural failure, calculating the non-linear tearing of the aluminum shell.

Hydrodynamic Scatter

Directional Velocity analysis X-Axis recorded the extreme dispersion of the fluid and debris upon rupture, with velocities ranging from 42.2 m/s to -65.7 m/s.



Business Value & Application

This dynamic FSI methodology is directly applicable to industrial crashworthiness testing, drop-test simulations for consumer electronics, and structural integrity assessments of pressurized tanks, drastically reducing the need for expensive physical prototyping.